V3.1 | Common Boat-Critical Components

ArduPilot Rover Survey Platform: Propulsion, Power, Navigation and Radio Links

Year 3 Engineering Group Project | Microplastic Mapping

5 November 2026 | Source-audited procurement concept, not a qualified boat

Scope rule. This document contains the **boat common core only**. Track A (live flow cell) and Track B (sample collection) have independent sensors, compute, pumping and payload budgets. The boat flight controller and the payload microcontroller exchange time, position, status and commands across a defined link. No payload hardware is counted in the boat cost.

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1 Requirements, design basis and procurement discipline

The supplied V3 Method Rationale defines a 20 m by 20 m area, a 120 min total mission and Gate G9 for adequate flotation and energy [1]. Those are **mission targets from the supplied project source**, not observed capabilities. The original ArduPilot boat tutorial is a *conceptual reference* rather than a component or weight specification [2]. The goal is a lighter differential-thrust twin-pontoon Rover, consistent with ArduPilot's boat and motor-configuration guides [3], [4].

Purchasing rule: every proposed purchasable item links to a registered supplier (Cytron or GI Electronic) or Shopee Malaysia; no manufacturer or overseas website is used as the purchase channel. Manufacturer/ArduPilot pages may be cited solely as *technical evidence*. Prices are RM displayed on 5 November 2026 or identified V2.4 source allowances. Seller variant prices, stock and cable connectors may change. **NR = not reported/verified for the sold SKU**; **A = allowance** (not a verified quote). The subtotal *cannot* be interpreted as a finished approved procurement quote.

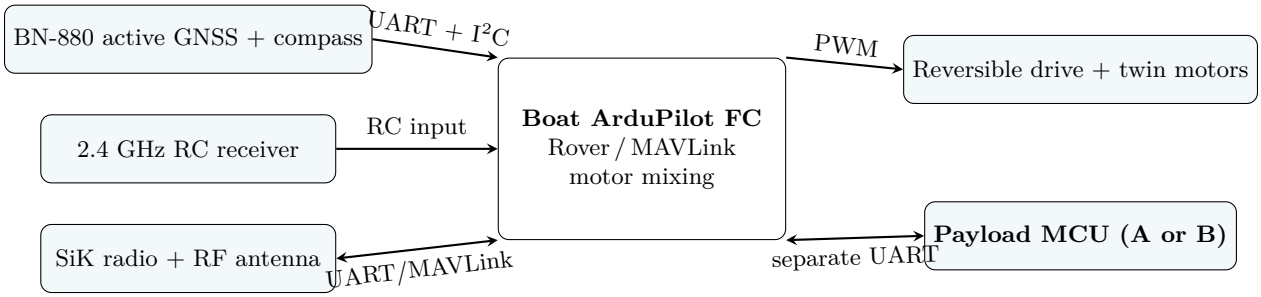
The overall budget constraint is **RM2,000**; the design should retain headroom instead of spending the entire ceiling. No 1–16 kg useful payload category is assigned to these configurations: physical payload mass and sensor power remain Track-dependent and must be separately measured. The COG, flotation, peak current, runtime and GNSS precision have not passed any physical test.

2 Antenna roles and communication architecture

Link	Purpose and component	What it does not do
GNSS receive	**Active GNSS patch/LNA** in Beitian BN-880 plus compass; receives satellite signals for positioning and heading calculations [5], [6], [7].	*Receive only:* cannot receive boat commands or send boat telemetry.
Manual RC	Dedicated **2.4 GHz** handheld transmitter + receiver, normally used for human take-over, mode switching and safety stop. HotRC bundle lists six channels and receiver included [8], [9].	RC channels are *not* automatically a Mission Planner MAVLink ground-station link.
Bidirectional RF	**SiK-style UART telemetry pair** with a boat antenna and ground antenna: carries MAVLink status and commands between Rover and a computer [10], [11], [12].	Not a substitute for the independent manual RC link until a tested fail-safe architecture has been approved.
Onboard Wi-Fi	CoreWing Mini V2 includes onboard Wi-Fi access-point/STA capability and supports Mission Planner TCP or UDP telemetry without a separate SiK transceiver [13].	A cost-saving *short-range development option*, not a verified outdoor communication link. No operating radius, waterproof antenna or drop-out performance is substantiated.

Cost-saving link variant. CoreWing Wi-Fi is already integrated in the budgeted flight controller [13]; omitting the unverified **RM160** SiK-pair allowance from Tier A reduces its *conditional* electronics-and-hull subtotal from RM1,192.37 to **RM1,032.37**, leaving **RM967.63** against the RM2,000 ceiling. This is an arithmetic planning comparison, *not* a tested, complete boat, and the independent 2.4 GHz RC link remains mandatory. The tier totals later retain SiK for comparison.

Active GPS detail. BN-880 is marketed with a patch antenna and compass; a BN-880 series technical sheet identifies an *active* antenna, but variant/genuine-module verification is required before purchase [5], [6], [7]. If a truly separate weatherproof GPS antenna is needed, GI Electronic's active antenna is available for **RM27**, claims 26 dB gain, 50 Ω impedance and 18 mA typical draw, and includes an adapter cable [14]. **Do not add it directly to BN-880:** an appropriate RF connector, GNSS receiver with active antenna bias voltage, water seal and cable-loss assessment are prerequisites; GI's antenna is an *alternative RF front end*, not an automatic performance upgrade.



UART allocation gate: dedicate one logical serial port to GNSS, one to SiK telemetry and one to payload MCU MAVLink, while RC uses its documented receiver interface. CoreWing's GPS, telemetry, external-compass and RC connectors are documented; **exact free UART pins, connector voltage and firmware build must be checked on the purchased board** [13]. MAVLink messages may be dropped, so synchronise timestamps and handle missing observations explicitly [15]. Use separate antenna positions and isolate high-current motor wiring from compass and RF leads.

3 Flight-controller comparison and SpeedyBee decision

ECU / purchase	Price, mass and power	Navigation / control evidence	Decision
CoreWing F405 Wing Mini V2 [16]	**RM264.43** voucher listing; 19 g board-family published figure; 2–6S [13], [16]	ArduPilot Rover board documentation lists GPS, compass, telemetry, receiver, motor outputs and integrated BEC; connector voltage and servo-rail BEC warnings apply [13].	**Preferred in tiers A–C.** UART/pin check required.
SpeedyBee F405 V4 **60A quad ESC stack** [17]	**RM445.40**; FC: 41.6 × 39.4 × 7.8 mm (10.5 g). ESC: 45.6 × 44 × 8 mm (23.5 g) [18]	FC has ArduPilot Rover support, 9 PWM outputs, 6 serial interfaces (some reserved), external I²C compass and microSD [18], [19].	**Comparator only.** ESC is quadcopter BLHeli_S; reversible boat operation *not confirmed.*

Why not buy the suggested stack first? The SpeedyBee ESC's 60 A channel rating is a *current limit* rather than a motor-power requirement, and its four-channel fixed-wing/quadcopter packaging creates no advantage for a two-motor boat. More importantly, ArduPilot's *bidirectional DShot telemetry* is **not the same** as physical forward/reverse propulsion; reversal depends on the ESC firmware/mode and motor protocol, and cannot be assumed from the published stack specification [4], [18], [19]. The stack may be retested later but is not costed as a proven reversing drive.

4 Propulsion and motor-driver technical comparison

Table 1: Motor and driver options: every rating is source-linked; unreported parameters are not inferred. Electrical input is NOT shaft output or horsepower.

Model/channel	**Supply**	**Dimensions**	**Mass**	**Thrust / shaft W / hp**	**Under-load power/current**	**Max current**	**Price each**	**Purchase; technical evidence**
12V three-blade jet unit, Shopee [20]	12V advertised	NR, sold variant	NR	thrust/shaft W/hp **NR**	current under water **NR**	stall/max **NR**	RM38.50–74.80 variant range; planning **74.80**	Shopee; no reliable maker sheet; motor type *unconfirmed*
Hobbyocean T60 + 40A ESC variations [21], [22]	12–16V and 2–4S *claimed by third-party spec*	NR for purchased SKU	NR	seller/third-party 300W electrical-rated class; thrust/hp for SKU **NR**	third-party T60 15–23A *claimed*; no certified curve	ESC 40A *variant claim*; motor limit NR	RM225.98–321.46 variants; **321.46 A**	Shopee; third-party tech, not procurement; verify *full kit*
M060 waterproof BLDC thruster [23], [24]	12–24V advertised	**NR exact SKU**	**NR exact SKU**	seller variant thrust/shaft W/hp **NR**; maker-family up-to-3kgf *not transferable*	max-loaded W/A exact SKU **NR**	max continuous/stall **NR**	**273.32**, voucher listing	Shopee; manufacturer-family reference
Cytron MDD10A, *brushed only* [25]	5–30V motor	84.5× 62 mm [25]	NR	n/a (H-bridge driver)	loss/heat case-specific	**10A continuous / channel**, check product manual; 2 channels	**108.78**	Cytron; NOT direct RC pulse input
Cytron MDD3A, *brushed only* [26]	4–16V motor	NR	NR	n/a	current load user-dependent	**3A continuous / channel, 5A peak**; 2 channels	**39.06**	Cytron; *not* safely sized without motor current
ZTW Shark G2 20A reversible ESC [27], [28]	2–4S advertised for 20A family	maker data for one variant; sold SKU NR	NR sold SKU	n/a; supports BLDC only	loss/thermal draw **NR**	20A *nominal selected variant*, actual motor matching unproven	**100.00 A**, seller listing RM67.65–227.70	Shopee; OEM family page

Motor-drive incompatibility gates. Tier A may use MDD10A *only after* a two-wire brushed motor is positively identified, its submerged load is measured below the channel limit and the controller input is translated into PWM/direction. BLDC motors require BLDC ESCs, not MDD10A. Tiers B and C require confirmed bidirectional 3S operation, independent PWM inputs, waterproof cabling, current limits and CW/CCW propeller arrangements. Voltage × current is *electrical input*; no horsepower figure can be honestly assigned without torque/rpm or shaft-dynamometer data.

5 Active GNSS, RC, telemetry and battery comparison

Table 2: Separate functions and technical evidence. Published values are attributed row by row.

Item	Electrical/physical	Interfaces, benefits and caveats	RM ea	Sourcing/evidence
Beitian BN-880 *active GNSS patch* + compass [5], [6], [7]	28× 28× 10 mm, approx 10 g; 3–5.5V family, 50mA capture typical (secondary docs)	Integrated active GNSS patch and I^2C compass claimed; serial GNSS output, compass on separate lines; vendor/board revision identification required	118.42 voucher	Shopee; maker; datasheet mirror
GI separate *active* GPS antenna [14]	26 dB gain; 50 Ω; 18 mA typical; 3m cable	SMA/IPX cable supplied; **requires compatible active-antenna RF receiver + bias**; *do not* connect to BN-880 by assumption	27.00	GI Electronic; receiver extra, not costed
M10 + IST8310 GNSS/compass [29]	exact antenna active/passive **NR**; mass/size NR	Alternate GNSS, but does *not* satisfy active-antenna requirement until antenna/LNA confirmed; cable pinout must be checked	125.09 voucher	Shopee
HotRC 6ch 2.4 GHz RC Tx + Rx [8]	Tx 230 g; receiver 4–9V; receiver mass NR	Six RC channels, fail-safe listed; packet protocol/receiver output interface confirm; as checked on 5 Nov backorder	109.00	Cytron, transmitter and receiver included
FlySky FS-i6X + receiver [30]	2.4GHz; size/mass exact chosen bundle NR	Alternative if HotRC unavailable; receiver *variant* and compatible iBUS/SBUS to be checked	207.98 display range example	Shopee; displayed entry may not include receiver
3DR/SiK **RF telemetry pair** + antennas [10], [12]	433/915MHz; listed 100mW–1000mW variants (*not* all selectable at minimum price)	MAVLink over UART to laptop ground radio; compatible TX/RX pair, frequencies and antennas must match; verify local RF rules	**160.00 A**	Shopee: 92.40–462 range, pair price NR; ArduPilot
Cytron 3S 2.2 Ah LiPo [31]	11.1V nominal; 30C advertised; 24.42Wh *derived*	Testing pack; at 80% usable, 120-min energy budget ≤9.77 W average *(calculation)*, not a runtime claim	75.00	Cytron
Cytron 3S 5.2 Ah LiPo [32]	11.1V nominal; 30C advertised; 145× 45× 28 mm; 57.72Wh *derived*	At 80% usable, 120-min energy budget ≤23.09 W average *(calculation)*; motor draws unknown	155.00	Cytron
Shopee XN Eagle 4S 14.8V 4200mAh LiPo [33]	14.8V nominal, 16.8V fully charged, 55C marketed; dimension and mass for 4S variant NR	62.16Wh nominal derived; at 80% usable, 120-min overall average ≤ 24.86 W (arithmetic only); 4S variant price NOT isolated from 2S–6S dropdown	174.67 *A*	Shopee; 4S exact SKU and inventory to confirm
Cytron B6AC 1–6S balancing charger [34]	100–240VAC or 11–18VDC; max 5A charge; 50W charge power; 575 g; 133 by 145 by 35 mm	Balance charges 4S pack on shore; configurable LiPo chemistry, terminal/lead type and safe current must be checked	95.00	Cytron; maker/distributor specs

Item	Envelope/power	Evidence	RM	Source
Cytron 2–3S balance charger [35]	110–240V AC, shore-only	Balanced charging, not boat payload; charger model and charging rate must match pack	19.90	Cytron

RF legal/integration gate: choose the 433 or 915MHz radio variant and configured transmit power only after checking MCMC/SIRIM conditions for intended field use; *no permission or licence exemption is asserted here.* The boat *may* receive waypoints or commands over MAVLink; ACK/HEARTBEAT, link drop-outs and failsafe behaviour need bench and shore trials [10], [11], [15]. **Manual override** must remain independent of MAVLink link availability.

6 Three increasing-cost common-core configurations

Voltage engineering correction. A 3S LiPo is only 11.1 V nominal, below the advertised 12V lower bound of the T60 and M060 thruster families. For tiers B/C, replace the original 3S 5.2 Ah pack with a **4S, 14.8V nominal, 4200mAh pack**; a Shopee XN Eagle multi-variant listing spans RM66.67–174.67, so **RM174.67 is an upper-range *allowance*, not a confirmed 4S price or stocked variant** [33]. Select the Cytron B6AC 1–6S compatible charger, **RM95** [34], since a 2–3S-only charger **cannot charge a 4S pack** [35]. *CRITICAL:* a fully charged 4S LiPo reaches 16.8 V; the T60 family sometimes quotes a 16V limit, so the motor supplier **must confirm full-charge voltage compatibility before 4S is connected** [21], [22]. This remains a design gate, not a proven electrical combination. For M060 12–24V, 4S nominal and charged voltage fall within its advertised range, subject to exact sold-SKU verification [23].

Important: These configurations do not vary by sample payload mass. All include two proposed thrusters, an ArduPilot-compatible boat FC, BN-880 *active* GNSS/compass, manual RC transmitter/receiver, a bidirectional RF telemetry pair, one LiPo, charger, pontoons, enclosure, fused wiring and isolation switch. The tier A motor/driver pairing is only a *conditional* study until brushed type and current are confirmed. Tier B pair pricing is a *variant-bounded allowance*, not a verified kit quotation. Tier C ESC prices are allowances. Cost inputs each have purchase references in the complete cost table below.

Tier	Twin motors / drive	FC, active GPS, radio	Main battery	Rough RM
A (low)	2 × generic 12V underwater jet (*brushed? NR*) + Cytron 2-ch MDD10A [20], [25]	CoreWing F405 Mini V2; BN-880 active; HotRC RC; 3DR/SiK pair (A) [5], [8], [12], [16]	3S 2200mAh [31]	**1192.37**
B (mid)	2 × T60-type BLDC + 40A reversible ESC bundles *only if verified* [21], [22]	Same common core and antenna/links [7], [8], [12], [16]	4S 4200mAh (A) [33]	**1751.68**
C (upper)	2 × M060 12–24V BLDC + 2 × ZTW 20A ESCs (A) [23], [27]	Same common core, GPS, RC and RF [5], [8], [12], [16]	4S 4200mAh (A) [33]	**1855.40**

Tier total *incl. basic hull*	Remaining vs RM2k	Share of cap	Budget judgement
A: **RM 1192.37**	RM 807.63	59.6%	Cheapest *research budget*, but NOT purchase-ready without identifying a brushed/compatible propulsion motor.
B: **RM 1751.68**	RM 248.32	87.6%	**Conditional only**: confirm T60 motor accepts 16.8V fully charged 4S, ESC variant, and physical forward/reverse.
C: **RM 1855.40**	RM 144.60	92.8%	Very limited remaining headroom; not preferred. Confirm exact motor/ESC variants, peak current and 4S charger.

4S budgeting caveat: Both higher tiers use an estimated 4S 4200mAh price at RM174.67 from a multi-variant Shopee listing. The exact 4S variant may cost less or more and may be unavailable. The old 3S/Cytron B3AC combination is not appropriate for a motor specified to require 12V minimum throughout discharge. [32], [33], [34],

[35]

Critical total caveat: These are planning subtotals from cited supplier pages and clearly marked allowances. Antenna mounts, correct wiring adapters, watertight fixing hardware, thruster brackets, RF approvals/shipping, propeller guards and final fabrication can change costs; battery and radio variants can change prices. The **payload MCU, sensors, pumps, filter/flow cell, sample bottles and image processing hardware are excluded**. The 2h gate is **not proven**. If the RM2,000 ceiling must include payload hardware too, a fresh combined budget is necessary once Track A/B is selected.

7 Source-audited itemised cost table (per configuration)

Table 3: Common-core cost basis: prices and assumptions attributed by IEEE reference on every row. A = allowance; voucher and variation prices can differ at checkout.

Tier	Purchasable item and source	Qty	RM ea	RM ext	Procurement status	Technical/sourcing caveat
A	CoreWing F405 Wing Mini V2 [16]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
A	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
A	HotRC 6ch Tx + Rx bundle [8]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
A	3DR/SiK air+ground radio pair [12]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
A	2-3S LiPo balance charger [35]	1	19.90	19.90	Published supplier price	Required only if not already available
A	IP65 box 263x182x60 mm [36]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion
A	110mm UPVC pipe 1m [37]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
A	110mm UPVC end cap [38]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked
A	PVC solvent cement [39]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
A	10-piece gland kit [40]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
A	Fuse holder and fuse [41]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
A	AWG18 by metre [42]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
A	12-48V battery isolator [43]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
A	12V 3-blade underwater jet thruster [20]	2	74.80	149.60	Upper visible variant / NOT MOTOR-SPECIFIC	Brushed/BLDC type, stall draw, CW and CCW pair unconfirmed
A	Cytron MDD10A 2ch brushed H-bridge [25]	1	108.78	108.78	Published supplier price	ONLY if motor is confirmed brushed and each channel <= 10A continuous
A	3S 11.1V 2200mAh LiPo [31]	1	75.00	75.00	Published supplier price	Short-duration bench/RC testing only; 2h not established
B	CoreWing F405 Wing Mini V2 [16]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
B	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
B	HotRC 6ch Tx + Rx bundle [8]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
B	3DR/SiK air+ground radio pair [12]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
B	Cytron 1–6S multifunction balance charger [34]	1	95.00	95.00	Published supplier price	4S-capable shore charger, matching connector required
B	IP65 box 263x182x60 mm [36]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion
B	110mm UPVC pipe 1m [37]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
B	110mm UPVC end cap [38]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked

Tier	Part/source	Qty	Unit	Ext	Price type	Constraint
B	PVC solvent cement [39]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
B	10-piece gland kit [40]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
B	Fuse holder and fuse [41]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
B	AWG18 by metre [42]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
B	12-48V battery isolator [43]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
B	Hobbyocean T60-type 16V BLDC + 40A ESC bundle [21]	2	321.46	642.92	Upper visible variant / BUNDLE UNCERTAIN	Seller variants may be motor-only/ESC-only; forward/reverse must be confirmed
B	4S 14.8V 4200mAh LiPo [33]	1	174.67	174.67	Upper listed variant / ALLOWANCE	Exact 4S price and inventory NR; 4S charged is 16.8V; verify thruster/ESC limits
C	CoreWing F405 Wing Mini V2 [16]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
C	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
C	HotRC 6ch Tx + Rx bundle [8]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
C	3DR/SiK air+ground radio pair [12]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
C	Cytron 1–6S multifunction balance charger [34]	1	95.00	95.00	Published supplier price	4S-capable shore charger, matching connector required
C	IP65 box 263x182x60 mm [36]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion
C	110mm UPVC pipe 1m [37]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
C	110mm UPVC end cap [38]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked
C	PVC solvent cement [39]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
C	10-piece gland kit [40]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
C	Fuse holder and fuse [41]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
C	AWG18 by metre [42]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
C	12-48V battery isolator [43]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
C	M060 underwater BLDC thruster 12-24V [23]	2	273.32	546.64	Observed voucher listing, 2 variants	No validated thruster input draw or thrust for exact variant
C	ZTW Shark G2 20A bidirectional ESC [27]	2	100.00	200.00	PLANNING ALLOWANCE	Listing RM67.65-227.70 over all variants; 20A exact not quoted; motor <=20A unproven
C	4S 14.8V 4200mAh LiPo [33]	1	174.67	174.67	Upper listed variant / ALLOWANCE	Exact 4S price and inventory NR; 4S charged is 16.8V; verify thruster/ESC limits

8 Mass, energy, buoyancy and validation gates

The hull concept reuses two 110 mm nominal-diameter 1 m UPVC pontoons, end-capped and solvent-bonded [37], [38], [39]. **Ideal external cylindrical displacement** (not verified load capacity) is $V_{\text{ext}} = 2\pi(0.055\,\text{m})^2(1.0\,\text{m}) = 19.0\,\text{L}$, calculated using the listed nominal diameter and nominal length; real waterplane, wall thickness, caps, freeboard and stability *reduce the usable margin*. At half immersion the simplistic submerged volume is approximately 9.5 L, so it is unsafe to infer 9.5 kg useful payload: *the entire boat mass* consumes this displacement. The V3 Method Rationale's stored-water scenarios cannot be assigned to this hull without a redesigned float geometry and direct flotation testing.

For 120-min operation, the new Tier B/C 4.2 Ah 4S pack has $14.8\,\text{V} \times 4.2\,\text{Ah} = 62.16\,\text{Wh}$ nominal (derived from the advertised cell count and capacity) [33]. If the *engineering assumption* is that only 80% of nominal energy is usable, the entire boat load must average $\leq 62.16 \times 0.8/2 \approx 24.86\,\text{W}$ to operate for 2 h, *without weather/current margin*. The Tier A 2.2 Ah 3S pack similarly provides only $11.1 \times 2.2 = 24.42\,\text{Wh}$ nominal, or $24.42 \times 0.8/2 = 9.77\,\text{W}$ over two hours [31]. These are **battery-only arithmetic constraints**, not measured endurance. Motor load curves and bridge efficiency remain unknown [20], [21], [23]. Two independent 3S packs are *not* assumed to be safely paralleled.

Pass/fail gates before any full-water trial:

1. **Procurement:** prove correct purchased motor CW/CCW variants, brush type, driver mode, ESC kit inclusion and exact checked-out price; verify RC receiver and air/ground radio pair include antennas.
2. **Dry electronics:** test 3S powered flight controller, active GNSS fix, compass away from thruster leads, independent RC disarm and MAVLink ground-station mode/waypoint command.
3. **Power protection:** check wire and fuse ratings; test battery isolator, BEC topology, common grounds, telemetry pins and potential ESC BEC backfeed.
4. **Waterproofing:** static leak test, float/freeboard and full-mass stability tests; test separately with each expected Track A/B payload installed, then assess turning and recovery in tethered water trials.
5. **Endurance:** log current at idle, straight cruise, turning and maximum thrust; measure usable battery Wh, then run 120 min with margin under representative conditions.
6. **Georeferencing:** confirm Boat FC sends timestamp, GPS fix status, latitude/longitude and movement state to the independent payload MCU over a dedicated serial MAVLink feed. Check actual lane-spacing errors in water.

9 Procurement order and final recommendation

Do not first purchase a SpeedyBee 60A stack or 4S battery before verifying motor voltage. Use the CoreWing controller, a BN-880 with active GNSS antenna and compass, a manual 2.4GHz RC transmitter/receiver and an RF SiK-style pair, subject to confirming available pins and radio frequency compliance [7], [8], [10], [13]. This provides an isolated navigation and communication backbone independent of whether V3 later selects Track A or Track B.

The most defensible *cost-versus-risk* option is **Tier B**, but only after the exact T60/ESC variants and forward/reverse command response are verified; otherwise retain **Tier C** as a fallback with proper M060 current measurement and ESC sizing. Tier A is a **low-cost proof-of-concept** only and must not be ordered until brushed motor identity, waterproofing and measured stall/underwater current are established. **No tier has validated 2h runtime or 1kg+ useful payload capacity.**

References (IEEE numbered source index)

[1] Y. 3. E. G. Project, *V3 method rationale: Microplastic survey boat*, Provided by project team as V3 Method Rationale archive, 2026.

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